

HARSHAVARDHAN K

Software Engineer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [LeetCode](#) | [HackerRank](#)

SUMMARY

Software Engineer with hands-on experience building embedded IoT systems, automation tools, and web-based applications using Python, JavaScript, and cloud platforms. Skilled at translating real-world problems into working technical solutions, demonstrated across academic projects and multi-institute hackathons. Comfortable across the stack — from sensor-level hardware integration to frontend interfaces and database design. Focused on writing clean, maintainable code and continuously strengthening problem-solving ability through competitive programming.

EDUCATION

B.Tech, Computer Science Engineering

Presidency University

CGPA: 8.8

2023 – 2027

Class XII

SDC College

93.83% (563/600)

2023

Class X

Vidhya Samskruthi International School

95.83%

2021

TECHNICAL SKILLS

Languages: Python, JavaScript

Web Technologies: HTML, CSS

Databases: SQL, MongoDB

Cloud: Google Cloud Platform (GCP)

Tools & Version Control: Git, GitHub, Visual Studio Code

PROJECTS

Smart City Tunnel Monitoring and Control System

[Python](#) | [Raspberry Pi Pico](#) | [RFID](#) | [OLED](#) | [Capacitive Touch Sensors](#)

- Engineered an embedded IoT monitoring system on Raspberry Pi Pico to track and manage conditions inside a tunnel environment in real time.
- Implemented RFID-based authentication to control access and strengthen security within the monitoring system.
- Developed an OLED display interface to surface real-time system status and sensor feedback to end users.
- Integrated capacitive touch sensors for contactless control input, improving responsiveness and system reliability.

Smart Watering System

[Arduino](#) | [Sensors](#) | [Relay](#) | [Water Pump](#)

- Designed an automated irrigation system using Arduino to monitor soil moisture levels and trigger watering cycles as needed.
- Programmed relay-controlled logic to switch a water pump on/off based on live sensor readings, reducing manual intervention.
- Built a closed-loop sensor-driven control system to optimize water usage and maintain consistent soil conditions.
- Applied embedded systems and automation principles to solve a practical, real-world use case.

INTERNSHIP

Web Development Intern — SkillCraft Technology

- Selected for a Web Development Internship focused on building and strengthening practical front-end and back-end development skills.

HACKATHONS & COMPETITIONS

- 3rd Place** — PALS Analyze Case Study, IT Engineering Category (44 Institutes), as part of Team Ignite Core, an IIT Madras initiative.
- Top 5** — Smart India Hackathon (Internal), Software Category, for the project *Arogya Sahaya Connect* — a portable healthcare records system.
- Participant** — PALS Tinkathon.

CERTIFICATIONS

JavaScript (Intermediate), SQL (Intermediate), CSS (Intermediate) — HackerRank

ADDITIONAL INFORMATION

Area of Interest

Frontend Development, Python Development, Cloud Computing, Machine Learning

Languages

English, Kannada, Telugu, Hindi

Hobbies

Problem Solving, Fitness, Badminton